

DynaKube Operator Deployment

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Installing and Configuring the Dynatrace Operator

The DynaKube operator is the recommended way to deploy Dynatrace monitoring in Kubernetes. This notebook covers installation via Helm, configuration options, and deployment modes for different use cases.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS tenant with API tokens
Kubernetes Cluster	v1.24+ with admin access
Helm	v3.x installed
kubectl	Configured for your cluster
API Tokens	Operator token + Data ingest token

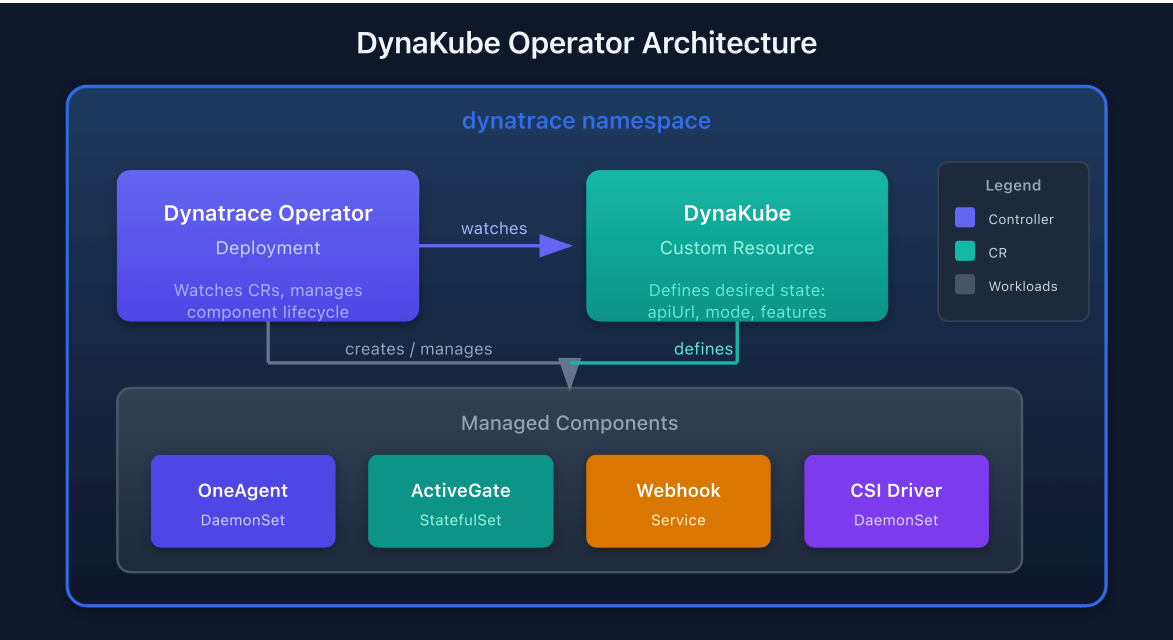
1. Operator Overview

The Dynatrace Operator manages the complete lifecycle of Dynatrace monitoring components.

What the Operator Manages

Component	Description	Managed By
OneAgent DaemonSet	Node-level monitoring	Operator
ActiveGate StatefulSet	Routing and K8s API access	Operator
Webhook	Code module injection	Operator
CSI Driver	Volume-based code modules	Operator

Operator Architecture



2. Prerequisites Setup

Required API Tokens

Create two tokens in Dynatrace with these scopes:

Operator Token:

Scope	Purpose
<code>activeGateTokenManagement.create</code>	ActiveGate tokens
<code>entities.read</code>	Entity information
<code>settings.read</code>	Read settings
<code>settings.write</code>	Write settings
<code>DataExport</code>	Export data (optional)
<code>InstallerDownload</code>	Download OneAgent

Data Ingest Token:

Scope	Purpose
<code>metrics.ingest</code>	Ingest metrics
<code>logs.ingest</code>	Ingest logs
<code>events.ingest</code>	Ingest events (optional)

Create Namespace and Secret

```
# Create namespace
kubectl create namespace dynatrace

# Create secret with tokens
kubectl create secret generic dynakube \
  --namespace dynatrace \
  --from-literal=apiToken=<OPERATOR_TOKEN> \
  --from-literal=dataIngestToken=<DATA_INGEST_TOKEN>
```

3. Helm Chart Installation

Add the Dynatrace Helm Repository

```
# Add Dynatrace Helm repo
helm repo add dynatrace
https://raw.githubusercontent.com/Dynatrace/dynatrace-
operator/main/config/helm/repos/stable

# Update repo cache
helm repo update

# View available versions
helm search repo dynatrace/dynatrace-operator --versions
```

Install the Operator

```
# Install operator only (DynaKube CR applied separately)
helm install dynatrace-operator dynatrace/dynatrace-operator \
  --namespace dynatrace \
  --create-namespace \
  --wait
```

Install with Custom Values

```
# Create values file
cat > dt-values.yaml <<< 'EOF'
# Operator configuration
operator:
  image: ""
  customPullSecret: ""

# Platform-specific settings
platform: "kubernetes" # or "openshift"

# Webhook configuration
webhook:
  hostNetwork: false

# CSI driver
csidriver:
  enabled: true
EOF

# Install with values
helm install dynatrace-operator dynatrace/dynatrace-operator \
  --namespace dynatrace \
  --values dt-values.yaml \
  --wait
```

4. DynaKube Custom Resource

The DynaKube CR defines how monitoring is deployed.

Minimal DynaKube (cloudNativeFullStack)

```
apiVersion: dynatrace.com/v1beta3
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  # Your Dynatrace environment URL
  apiUrl: https://ENVIRONMENT_ID.live.dynatrace.com/api

  # OneAgent deployment mode
  oneAgent:
    cloudNativeFullStack:
      tolerations:
        - effect: NoSchedule
          key: node-role.kubernetes.io/master
          operator: Exists
        - effect: NoSchedule
          key: node-role.kubernetes.io/control-plane
          operator: Exists

  # ActiveGate for Kubernetes monitoring
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
      - dynatrace-api
```

Apply the DynaKube

```
# Apply DynaKube CR
kubectl apply -f dynakube.yaml

# Watch deployment progress
kubectl -n dynatrace get dynakube -w
```

5. Deployment Modes Explained

cloudNativeFullStack (Recommended)

```
oneAgent:
  cloudNativeFullStack:
    # Injected via webhook using CSI driver volumes
```

Pros	Cons
No privileged containers for apps	Requires CSI driver
Best for multi-tenant clusters	Slightly more complex
Independent app/infra monitoring	

classicFullStack

```
oneAgent:
  classicFullStack:
    # OneAgent DaemonSet provides code modules via host mounts
```

Pros	Cons
Simpler architecture	All pods share OneAgent
Single component	Requires hostPath mounts

applicationMonitoring

```
oneAgent:
  applicationMonitoring:
    # Only application-level monitoring, no infrastructure
    useCSIDriver: true
```

Pros	Cons
Minimal footprint	No infrastructure visibility
No node-level access needed	No container metrics
Good for managed K8s	

hostMonitoring

```
oneAgent:
  hostMonitoring: {}
```

Pros	Cons
Infrastructure only	No application tracing
Lightweight	No code-level insight

6. Configuration Options

Resource Limits

```
spec:
  oneAgent:
    cloudNativeFullStack:
      resources:
        requests:
          cpu: 100m
          memory: 256Mi
        limits:
          cpu: 300m
          memory: 512Mi
```


Node Selectors and Tolerations

```
spec:
  oneAgent:
    cloudNativeFullStack:
      nodeSelector:
        kubernetes.io/os: linux
      tolerations:
        - key: "dedicated"
          operator: "Equal"
          value: "monitoring"
          effect: "NoSchedule"
```

Namespace Selectors

Control which namespaces get injection:

```
spec:
  namespaceSelector:
    matchLabels:
      dynatrace-injection: enabled
```

Or exclude specific namespaces:

```
spec:
  namespaceSelector:
    matchExpressions:
      - key: dynatrace-injection
        operator: NotIn
        values:
          - disabled
```

Custom ActiveGate Configuration

```
spec:
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
    replicas: 2
    resources:
      requests:
        cpu: 500m
        memory: 512Mi
      limits:
        cpu: 1000m
        memory: 1Gi
    topologySpreadConstraints:
      - maxSkew: 1
        topologyKey: topology.kubernetes.io/zone
        whenUnsatisfiable: ScheduleAnyway
```

Feature Flags and Annotations

Control injection at the pod level:

Annotation	Effect
oneagent.dynatrace.com/inject: "false"	Disable injection for pod
oneagent.dynatrace.com/inject: "true"	Force enable injection
oneagent.dynatrace.com/technologies: "java,nodejs"	Limit technologies

DynaKube feature flags:

```
spec:
  oneAgent:
    cloudNativeFullStack:
      env:
        - name: ONEAGENT_ENABLE_VOLUME_STORAGE
          value: "true"
```

Templates Configuration

Configure OTel Collector and Log Monitoring via `spec.templates` :

```
spec:
  templates:
    otelCollector:
      # ⚠ Always pin to specific version – avoid 'latest'
      imageRef:
        repository: public.ecr.aws/dynatrace/dynatrace-otel-collector
        tag: "0.16.0" # Check releases for current version
      resources:
        requests:
          cpu: 100m
          memory: 256Mi
        limits:
          cpu: 500m
          memory: 512Mi

    logMonitoring:
      resources:
        requests:
          cpu: 100m
          memory: 256Mi
        limits:
          cpu: 500m
          memory: 512Mi
      tolerations:
        - effect: NoSchedule
          key: node-role.kubernetes.io/control-plane
          operator: Exists
```

Note: Log monitoring is enabled with `spec.logMonitoring: {}` but resource configuration goes under `spec.templates.logMonitoring` .

OTel Collector and Log Monitoring Sizing Guide

Environment	CPU Limit	Memory Limit	Notes
Dev/Test	200m	256Mi	Low telemetry volume
Staging	500m	512Mi	Medium volume
Production	1000m	1Gi	High volume

7. Verification and Validation

Check Operator Status

```
# Operator pods
kubectl -n dynatrace get pods -l app.kubernetes.io/name=dynatrace-operator

# DynaKube status
kubectl -n dynatrace get dynakube

# Detailed status
kubectl -n dynatrace describe dynakube dynakube
```

Check OneAgent Deployment

```
# OneAgent DaemonSet
kubectl -n dynatrace get daemonset -l app.kubernetes.io/component=oneagent

# OneAgent pods on each node
kubectl -n dynatrace get pods -l app.kubernetes.io/component=oneagent -o wide
```

Check ActiveGate

```
# ActiveGate StatefulSet
kubectl -n dynatrace get statefulset -l
app.kubernetes.io/component=activegate

# ActiveGate pods
kubectl -n dynatrace get pods -l app.kubernetes.io/component=activegate
```

DynaKube Status Phases

Phase	Meaning
Running	All components healthy
Deploying	Components being created
Error	Check events for details

```
// Verify Kubernetes cluster is reporting to Dynatrace
fetch dt.entity.kubernetes_cluster
| fields entity.name, tags
| sort entity.name asc
```

```
// Check OneAgent deployment health via events
fetch logs, from:-1h
| filter matchesPhrase(content, "dynatrace") and matchesPhrase(content,
"oneagent")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

8. Upgrading the Operator

Helm Upgrade Process

```
# Update Helm repo
helm repo update

# Check current version
helm list -n dynatrace

# Upgrade to latest
helm upgrade dynatrace-operator dynatrace/dynatrace-operator \
  --namespace dynatrace \
  --reuse-values \
  --wait
```

Version Compatibility

Operator Version	Min K8s	Max K8s	Notes
1.0.x	1.21	1.28	Legacy
1.1.x	1.23	1.29	Current
1.2.x	1.24	1.30	Latest

Rollback if Needed

```
# View history
helm history dynatrace-operator -n dynatrace

# Rollback to previous version
helm rollback dynatrace-operator 1 -n dynatrace
```

Next Steps

Now that the operator is deployed, proceed to:

Next Notebook	Topic
K8S-03: GitOps for DynaKube	Manage DynaKube with ArgoCD/Flux
K8S-04: Cluster Health Monitoring	Deep-dive into cluster metrics
K8S-05: Workload Monitoring	Application-level observability

Summary

In this notebook, you learned:

- Dynatrace Operator architecture and components
- Prerequisites: API tokens and namespace setup
- Helm chart installation with custom values

- DynaKube CR structure and configuration
 - Deployment modes and when to use each
 - Configuration options for resources, selectors, and features
 - Verification commands and status phases
 - Upgrade and rollback procedures
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References

- [Dynatrace Operator](#)
 - [DynaKube CRD Reference](#)
 - [Helm Chart Values](#)
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